

Application of Water Quality Index (WQI) in Some Waterbodies in Enugu-Agidi, Njikoka LGA of Anambra State, Southeastern Nigeria

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Abstract

Water is a very important resource, especially for its usage in domestic and irrigation purposes. It is therefore pertinent to evaluate and determine the quality of water in our locality. In this study, five (5) freshwater bodies, namely Obibia River, Ogbunago Stream, Ovia Stream, Otukpuma River, and Diochiogba River, located at Enugu-Agidi, Njikoka LGA of Anambra State, were evaluated for their suitability for drinking and other domestic use. Three water samples were collected in-situ from each stream, making it a total of fifteen water samples from the five sites, and analyzed in the laboratory for their physicochemical and heavy metal content. Thirteen (13) physicochemical parameters and four (4) heavy metals were used to determine the suitability of the water based on WHO 2022 addendum standards. The physicochemical parameters include: pH, TDS, EC, hardness, temperature, chloride, nitrate, BOD, DO, color, odor, appearance, and turbidity. The heavy metals include iron, arsenic, zinc, and lead. The heavy metals within the water body are spatially analyzed using a bar chart, which shows that iron has the highest concentration, followed by zinc, and the lowest is arsenic. The Pearson correlation analysis was also used to unravel the complex relationship of the 15 water quality parameters examined across five sampling points. In the Pearson correlation matrix, strong positive correlation was observed in conductivity, turbidity, nitrate, and iron, with values ranging from 0.84 to 0.98. Conductivity correlated 0.98 with iron, 0.89 with turbidity, and 0.84 with nitrate. Similarly, turbidity showed a correlation of 0.92 with nitrate. Temperature exhibits a strong negative relationship with pH (-0.73), hardness (-0.57), and lead (-0.91). Lead (Pb) has a strong negative correlation with temperature, which means lead solubility and mobility increase substantially with decreasing temperature. The principal component analysis (PCA) was also used to explain the variance, with PC1 explaining 33.15% of the total variance to represent conductivity, turbidity, nitrate, and biological oxygen demand. PC2 similarly explains 30.56% of the variance controlled by pH and temperature. PC3 represents 22% of the mineralization and salinity concentration, and PC4, 13.29%, explains the variation in the component that captures arsenic and lead. Whereas PC5 measures the noise and is less than 1%. Therefore, the physicochemical parameters and heavy metals were compared to the WHO standard, and the general water quality index was greater than 100 (WQI value > 100) for the five rivers; when compared with BIS-WQI (Bureau of Indian Standards Water Quality Index), the value was greater than 300 (WQI value > 300), which, when placed on their classification scheme, shows that the water was unsuitable for drinking and other uses. Therefore, from the results obtained in the statistical analysis, the rural community using this water should be cautious of infection and other diseases, as it is recommended that government and non-governmental agencies be involved in producing suitable and palatable drinking water for the community affected that solely relies on this stream and river for their daily survival.

Keywords: Physicochemical; Water; Heavy Metal; Statistical Analysis; Principal Component.

1. Introduction

Groundwater is a very critical component of the ecosystem, providing about 65% of drinking water, 20% for irrigation purposes, and the remaining 15% for industrial use (Adimalla et al., 2018), and its quality usually revolves around the physical, chemical, and biological components of the waterbody in relation to its suitability for a particular use (Panagiotou et al., 2024; Rhishi Hari Raj et al., 2025). The quality of water has been declining as a result of anthropogenic and natural events within the environment (Oualid et al., 2025; Algretawee et al., 2021). This is as a result of human activities that greatly influence the quality of water within a given vicinity, which include intensive

agricultural activities, mining, industrialization, and urbanization that lead to unplanned disposal of their waste into the surface water body (Pareta et al., 2024; Edokpayi et al., 2017). Also, the changes in land use and land cover caused by increased sediment runoff and soil erosion can lead to serious water degradation, which will definitely disrupt the ecosystem (Ez-zaouy et al., 2025; Cantillana et al., 2024; Al-Shammari et al., 2020; Lobato et al., 2015). Furthermore, *Escherichia coli* (*E. coli*) contamination in surface water bodies has become a growing concern as a result of improper human feces deposition and the insanitary condition of the riverbank, which sprays around the waterbody and concentrates mainly around its source. All these external processes that lead to surface water contamination strongly affect the chemical composition of water, which is largely controlled by hydrogeological processes operating within the waterbody. These processes include evaporation, mineral dissolution, ion exchange, oxidation-reduction reactions, and groundwater mixing, which in turn influence the concentration and distribution of the major ions and trace elements (Akoto 2022). According to the World Health Organization (WHO) 2011, 80% of polluted water is the major cause of human diseases worldwide, which must be curtailed in order to have a healthy environment that supports good living. We have different indicators that are widely used in evaluating water quality, including pH, which indicates the acidity or alkalinity of water. The WHO always recommends a pH range of 6.5–8.5 for drinking water (WHO, 2022). Very high pH values usually alter the solubility of metals and, at the same time, affect aquatic organisms. Total Dissolved Solids (TDS) and conductivity reflect the concentration of dissolved ions. Excessive TDS (>500 mg/L) in water reduces the palatability for drinking and can affect the soil quality, especially during irrigation (Palamuleni and Akoth, 2015). Dissolved oxygen is one of the essential physical parameters that support most of the aquatic organisms in the water body. Its concentration below 5 mg/L is considered suitable and supports life (Ogah et al., 2025; and Tyagi et al., 2013). Biological Oxygen Demand (BOD) represents the oxygen required by microorganisms to decompose organic matter. High BOD (>6 mg/L) most often signals organic pollution and low water quality (Yisa and Jimoh, 2010). The clarity of a water sample is usually associated with turbidity. Turbidity measures the amount of suspended material in a water body, and when its influence is strong, it prevents photosynthesis processes and harbors pathogen culturing. The permissible limit by WHO 2022 for turbidity is 5 NTU. Heavy metals are the major causes of pollution that threaten life due to their persistence, bioaccumulation, and toxicity to human and aquatic life. Heavy metals such as cobalt (Co), cadmium (Cd), copper (Cu), zinc (Zn), and chromium (Cr) play an important role in human metabolism at a minimal trace level but become excessively harmful when exposed to high concentrations (Abedi et al., 2019). Lead and arsenic are especially harmful, with safe limits of 0.01 mg/L and 0.01 mg/L, respectively, for drinking water (WHO, 2017). Groundwater reports in parts of Bayelsa and Delta, specifically Yenagoa and Warri respectively, show high levels of Pb, Fe, Cd, and Cr, particularly during the dry season and in trace quantities during the rainy season (Chinemelu et al., 2022; Eguakun et al., 2020). Also, in Obio Akpor in Rivers State, the concentration of lead (Pb) is high, and toward the Elele axis, it is within their permissible standard for drinking, as reported by (Offodile et al. (2022; Oborie et al. 2017); furthermore, the Calabar South Local Government Area shows heavy metal concentrations to be within the permissible limits for consumption. (Njar et al., 2012).

Therefore, all the parameters used for water in the evaluation of the water quality index provide critical insights into both the natural conditions and human-induced impacts on rivers. When evaluated together, they offer a holistic view of water quality status. The Water Quality Index (WQI) is a composite indicator that is simplified to interpret multiple water quality parameters into a single numerical value. The simplification of vast, complex water quality datasets was first introduced by Horton in 1965. Its foundation was aimed at providing a simple representation of water quality. It has since been widely applied in water resource sustainability and management worldwide for various water quality assessments, especially for drinking, wastewater, recreational, and irrigational purposes. Also, Shahri and Zaline (2025) strategize and explain the analysis of water quality using a bibliometric analysis of publications on the water quality index from 1973 to 2024, explaining the wide range of WQI application and usage in research and practical application worldwide. We have different WQI models, which include NSFQI (National Sanitation Foundation Water Quality Index), which was developed in 1970 and it uses parameters such as DO, BOD, fecal coliform, nitrate, and turbidity. The Canadian Council of Ministers of the Environment Water (CCME) has developed a globally recognized model that has helped in water quality assessment; the result of the model has shown 100% accuracy, as discussed by Swain (2025). Swain 2025 states that the model focuses on the magnitude and frequency of exceedance of the limit that is accepted for drinking water, and this makes it viable to be used across different regions of the globe. The Weighted Arithmetic Index model is commonly applied in developing countries; it assigns weights to parameters based on their relative importance and compares observed values against permissible standards (Tyagi et al., 2013). WQI transforms complex datasets into a single, comprehensible value (Sutadian et al., 2016). It also facilitates communication of water quality status to policymakers and communities, which enables spatial and temporal comparisons of water bodies, providing decision support for sustainable water management. WQI has been applied extensively across different places, countries, and continents around the globe. Other countries of the world have used different statistical approaches for water quality assessment, notably in India, where the study of the water quality of the Ganges and Yamuna River, revealed poor water quality as a result of industrial effluent and sewage from industries. Also in China, WQI analysis of the Yangtze River was conducted to ascertain the level of its suitability for drinking, and the results indicate variability in contamination through rainfall and surface runoff (Wang et al., 2017). In Canada, CCME WQI has been institutionalized for nationwide monitoring programs and societies. African studies also have similar challenges; notably, in Nigeria, studies have highlighted the deteriorating water quality in rivers and streams. According to Akoteyon et al. (2011) rivers in Lagos State were heavily polluted due to urbanization and human activity, with the water quality index value indicating very poor quality for use. Also, Egbueri (2018) assessed groundwater in different regions in Anambra State using WQI and reported contamination from heavy metals and anthropogenic activities. Ogbiebu et al. (2014) and Edegbai et al., (2026) evaluated rivers in southern Nigeria and found out that most were unsuitable for drinking, though some were usable for irrigation. In Ethiopia, WQI was also applied to rivers near Addis Ababa and reported severe organic pollution. In Ghana, WQI analysis of the Volta Basin also showed that while water was suitable for irrigation, it was unsafe for drinking due to microbial and chemical contamination (Oketola and Ad-ekolurejo, 2010). Studies all over the world have successfully used these tools in assessing the quality of water for its usage to ascertain the degradation causes to health and agricultural product yield, notably the works of Vijayakumar et al. (2025); Machona et al. (2025); Meghanad et al. (2025); Rahman et al. (2025); Hemalatha et al. (2025); Onana et al. (2025); and Chaudhary et al. (2024). This study is important in providing continuous monitoring of rivers/streams in our local region and also contributes to the research community on the need for appropriate WQI to ensure safe and suitable groundwater use, particularly in regions that are facing contamination through anthropogenic activity and natural events. According to Gupta and Gupta (2021), the water quality index (WQI) assessment from 2010 to 2019 was reviewed, and it highlights the importance and limitation of using this method in assessing the quality of water within a given vicinity.

Despite the level of awareness of water contamination, whether high or low, the evaluation of the water quality index is critically essential for good health and community development. Many small rivers and streams in Enugu-Agidi, such as the Obibia, Ogbunago, Ovia, Otukpuma, and Diochiogba streams, that are studied in this research, are directly relied upon by rural communities but lack consistent and effective evaluation. There is a lack of comprehensive datasets on these rivers, which limits policy development and community awareness. Thus, applying WQI to these water bodies bridges an important knowledge gap and supports evidence-based water resource management.

2. Description of The Study Area

The study focused on five freshwater bodies: Obibia River, Ogbunago Stream, Ovia Stream, Otukpuma River, and Diochiogba River. These rivers and streams are located within longitude ($6^{\circ}58'40''$ E - $7^{\circ}02'10''$ E) and latitude ($6^{\circ}11'30''$ N - $6^{\circ}13'50''$ N) in southeastern Nigeria, precisely at Enugu-Agidi, Njikoka LGA of Anambra State. The study location map (Figure 1) shows the topographic distribution within the study area and region where the samples of the water body were collected. The map displays the topographic features of the study area, with lowlands ranging from 47 to 93m where the streams are mostly located, and the highlands ranging from 191 to 267m. The area shares a boundary with Kogi to the north, Imo to the south, Enugu to the east, and Delta to the west and is characterized by rapid population growth, urban expansion, and agricultural activities.

Geologically, the area falls within the Anambra Basin in the Lower Benue Trough tectonic unit, which is one of the sedimentary basins in SE Nigeria. The Benue Trough was formed due to rifting during the breakup of Gondwana and the opening of the South Atlantic Ocean. The trough experiences a cycle of subsidence, sedimentation, marine transgression, and compressional tectonics, which led to the deposition of the sediments in the area. This has been studied by many researchers, including Wright (1968). Sediments in the basin were deposited during the Cretaceous age, with the bedding striking in the direction of NE-SW and dipping gently towards the northern direction.

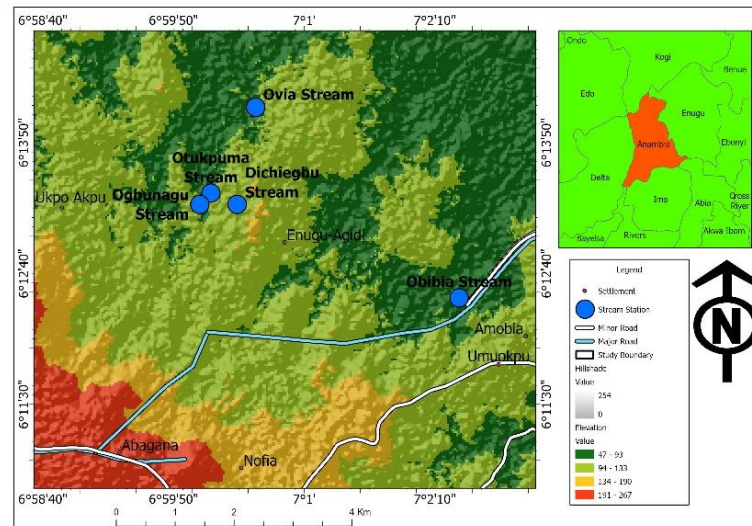


Fig. 1: Study Location Map of the Area Showing the Topographic and Stream Location.

3. Methodology

3.1. Sampling

A total of fifteen (15) water samples were collected in situ from the five (5) different rivers in clean and sterilized plastic bottles. Three samples were collected per site to ensure representativeness, making it a total of fifteen primary samples for laboratory analysis. Sampling was conducted during the wet season in May 2023. The sampling bottles were initially placed in a nitric acid solution of about 5% for 24 hours, rinsed with deionized water, and finally rinsed again with the river's water. Samples for heavy metal analysis were acidified with nitric acid to a pH below 2 immediately after collection to prevent precipitation or adsorption in the container walls. All samples were transported to the laboratory in a safe pack to maintain a temperature below 4°C (Singh et al., 2005), thereby limiting biological activity until the analysis was carried out. The following physicochemical and heavy metal parameters were measured: pH (acidity/alkalinity), electrical conductivity ($\mu\text{S}/\text{cm}$), total dissolved solids (mg/L), temperature ($^{\circ}\text{C}$), turbidity (NTU), nitrate (mg/L), total hardness (mg/L CaCO_3), chloride (mg/L), dissolved oxygen (DO, mg/L), biochemical oxygen demand (BOD, mg/L), and metals: iron (Fe), zinc (Zn), lead (Pb), and arsenic (As).

3.2. Analysis

The analytical procedures followed standard methods for the examination of water and wastewater for quality (APHA, 1998; APHA 2005; APHA 2017). The pH value and temperature of the sample were measured in situ with the aid of a digital pH meter (Hanna HI991300) and thermometer, respectively. Total dissolved solids (TDS) were estimated gravimetrically by evaporating a known volume of filtered sample at 180°C ; dissolved oxygen (DO), biochemical oxygen demand (BOD_5), nitrate (NO_3^-), chloride, hardness, and turbidity were measured and estimated by standard methods (APHA 2017). Then heavy metals (Fe, Zn, Pb, As) were analyzed using Atomic Absorption Spectrophotometry (AAS, Varian AA240) after acid digestion of water samples. Data were analyzed using both descriptive and inferential statistical techniques. Descriptive statistics such as mean, minimum value, and maximum value were computed to show the variability of each parameter across the rivers. The computed value was executed in an Excel spreadsheet and compared against the WHO (2022) standard for water quality guidelines to assess compliance. The calculated WQI values were used to classify the rivers by suitability for domestic and agricultural uses. In addition, heavy metal concentrations were plotted using a bar chart to show the distribution of the concentration in each water body, which will aid in the identification of streams with high metal content. Furthermore, the correlation plot of the measured and estimated parameters was computed to determine the relationship between each parameter and the others in the study. Statistical analyses and graphical representations were carried out using Microsoft Excel, ArcGIS Pro, and Anaconda.

4. Results and Discussion

The results of the physicochemical properties of the five rivers in the study area are presented in Table 1. The results (Table 1) present the concentration of the physicochemical parameters of the rivers considered for the analysis. The organoleptic properties show that all the water samples in the stream and rivers were objectionable in appearance and odor, with the exception of the Ovia stream. The statistical analysis of the various parameters, such as the maximum, minimum, and mean/average value, was also outlined in order to show the variation in each parameter.

Table 1: The Physicochemical Concentration and Statistical Analysis of the Five Rivers in the Study Area

Parameter	Otukpuma Rivers	Obibia River	Ogbunago Stream	Ovia Stream	Diochiogba Rivers	Max	Min	Average	Standard (WHO 2022)
EC ($\mu\text{S}/\text{Cm}$)	136.00	120.00	128.00	110.00	115.00	136.00	110.00	121.80	1000.00
TDS (Mg/L)	32.00	56.00	33.00	32.00	51.00	56.00	32.00	40.80	500.00
PH	7.40	7.66	7.52	7.58	4.74	7.66	4.74	6.98	7.00
Temperature ($^{\circ}\text{C}$)	30.80	30.00	30.00	30.40	31.00	31.00	30.00	30.44	25.00
Turbidity (NTU)	2.40	0.50	0.80	0.30	0.40	2.40	0.30	0.88	5.00
Color(pt.co)	3.00	5.00	4.00	2.00	4.00	5.00	2.00	3.60	15.00
Hardness (Mg/l)	2100.00	2100.00	1800.00	2000.00	2100.00	2100.00	1800.00	2020.00	300.00
BOD(Mg/l)	64.00	70.00	105.00	78.00	100.00	105.00	64.00	83.40	5.00
Dissolved oxygen (Mg/l)	16.40	17.20	19.40	15.20	17.80	19.40	15.20	17.20	5.00
Nitrate (Mg/l)	0.39	0.23	0.16	0.10	0.11	0.39	0.10	0.20	50.00
Chloride (Mg/l)	122.00	138.00	190.00	120.00	135.00	190.00	120.00	141.00	250.00
Lead (Mg/l)	0.10	0.14	0.17	0.12	0.10	0.17	0.10	0.13	0.01
Iron (Mg/l)	0.50	0.28	0.39	0.20	0.28	0.50	0.20	0.33	0.30
Zinc (Mg/l)	0.38	0.40	0.32	0.30	0.42	0.42	0.30	0.36	3.00
Arsenic (Mg/l)	0.01	0.02	0.01	0.02	0.01	0.02	0.01	0.01	0.01
Odor	Obj	Obj	Obj	UnObj	Obj	-	-	-	-
Appearance	Obj	Obj	Obj	UnObj	Obj	-	-	-	-

Where: Obj-objectionable, UnObj-unobjectionable

Seventeen (17) physicochemical parameters were analyzed from the five water bodies. The average value of each of the concentrations was compared with the WHO-2022 permissible standard; it was observed that the majority of the parameters (Zn, Cl⁻, NO₃⁻, color, turbidity, EC, TDS, and pH) were within the permissible standard for drinking water, whereas the rest were above the drinkable standard. The increase in dissolved oxygen in the water body above the limit shows the availability of aquatic animals in the environment and is very important. Therefore, the presence of dissolved oxygen plays a critical part in the survival and composition of the ecosystem. The dissolved oxygen in the individual water body was above the WHO 2022 standard, and the maximum value was observed in Ogbunago Stream, which is a critical parameter in the survival of fish in the area and other aquatic organisms. Most of the heavy metals in the region are contributing factors to the pollution of water quality in the region, since these might be gotten from the weathering and erosion of rocks within the study area. When comparing the concentration of the heavy metals with respect to their particular location or water body, as seen in Figure 2. The presence of iron (Fe) is the most dominant, and arsenic is the least dominant in terms of concentration. Iron presence in water may be because of corrosion of materials, and is usually not considered one of the harmful parameters for water quality. The presence of zinc and lead in the water body is the major contaminant and pollutant that originates from the weathering of sphalerite and galena, respectively, in the study area.

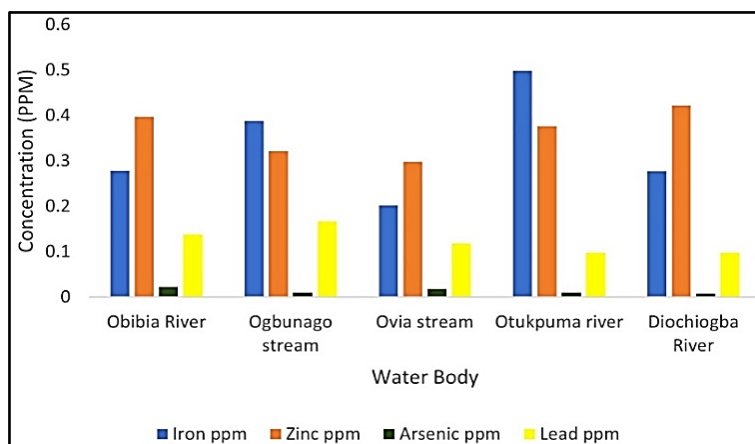


Fig. 2: Pictorial Representation of the Heavy Metal in the Water Body.

4.1. Correlation matrix of water quality parameters

The statistical analysis and description of the physicochemical properties of the water body were tested through the correlation matrix in Figure 3, which shows the degree of linear association between different water quality parameters, with values ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation). The Pearson correlation analysis unravels the complex relationship of the fifteen (15) water quality parameters examined across five sampling points.

In the matrix, the positive and negative values give in-depth knowledge about the pollution sources, chemical equilibria, and parameter redundancy in the water quality assessment. Strong positive correlations were observed among conductivity, turbidity, nitrate, and iron, with correlations ranging from 0.84 to 0.98. Conductivity exhibits 0.98 with iron, 0.89 with turbidity, and 0.84 with nitrate. Similarly, turbidity showed a correlation of 0.92 with nitrate. This high interrelationship between these parameters indicates high pollution sources that are linked to the environment. According to Singh et al. (2005), this high cluster indicates the conductivity of the water with respect to dissolved materials, which may increase dissolved ionic species. The finding is similar to the observation by Ouyang (2005) in a river system affected by mixed land use when sediment runoff carries both nutrients and metals from industrial and metallic materials. Nitrate and turbidity usually come from agricultural residue. Chloride is a conservative ion that indicates groundwater contribution or saline intrusion. Temperature exhibits a strong negative relationship with pH (-0.73), hardness (-0.57), and lead (-0.91). Lead has a strong negative correlation with temperature, which means lead solubility and mobility increase substantially with decreasing temperature. This could alternatively mean lead and organic matter precipitate at high temperatures (Forstner and Wittmann, 2012).

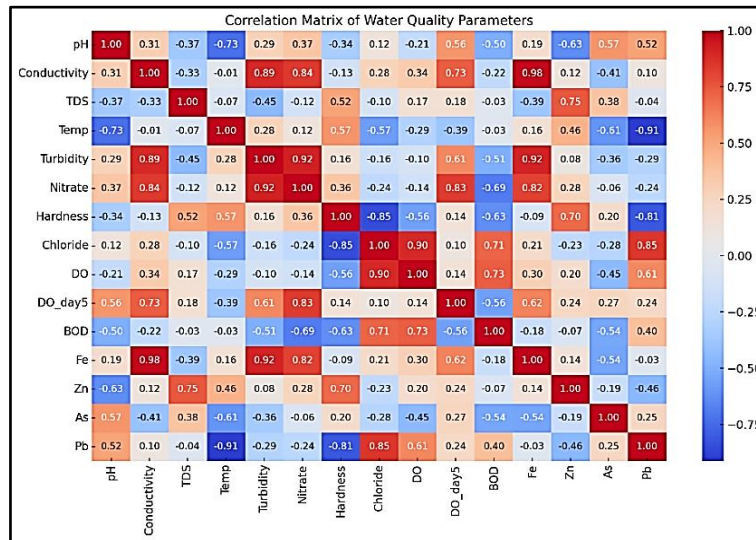


Fig. 3: Pearson Correlation Plot of the Physicochemical Parameters Obtained from the Water Samples.

4.2. Principal component analysis (PCA)

Principal component analysis tools are useful in reducing the fifteen (15) parameters while retaining their original meaning and interpretation. The principal component analysis is presented in Fig. 4, showing the description of the parameters in a scree plot, which explains the variance of the dataset. The scree plots explain that the first four principal components capture the meaningful variance in the dataset, with the fifth component representing noise.

The summation of the five variances ($PC1+PC2+PC3+PC4+PC5$) = 100%; PC1 and PC2 alone explain about 63.71% of the total variance. PC1 explains 33.15% of the total variance and represents conductivity, turbidity, nitrate, and biochemical oxygen demand. It summarizes the surface influence, pollution and anthropogenic inputs, organic matter, dissolved ions, variability of industrial effluent, and agricultural runoff on water quality. The results of geochemical variation presented in Table 2 show that, for the factors in PC1, only BOD was above the WHO 2022 permissible standard for drinking water, which means that the five water samples it contains a larger amount of biodegradable organic matter, which may come from sewage, wastewater, agricultural runoff, decaying vegetation, or other organic wastes. PC2 similarly explains 30.56% of the variance controlled by pH and temperature. It helps in understanding the seasonal variation in the water body as a result of fluctuations, hydrochemical equilibrium, water-rock interaction, environmental thermal conditions, and mixing of dissolved materials and changes in metal behavior, which result in the amount of inorganic and organic materials that are dissolved and mix with the water body. PC3 represents 22% of the mineralization and salinity concentration. It shows the relationship between dissolution, ionic enrichment, evaporation, concentration of ionic components, chloride, hardness, and dissolved oxygen. PC4 represents the variation in the component that captures arsenic and lead in other hand, explaining the trace metal enrichment and major sources of the contaminants. In this study, this is traced down to geogenic or anthropogenic influence on the water body. Whereas PC5 measures the noise and is less than 1%.

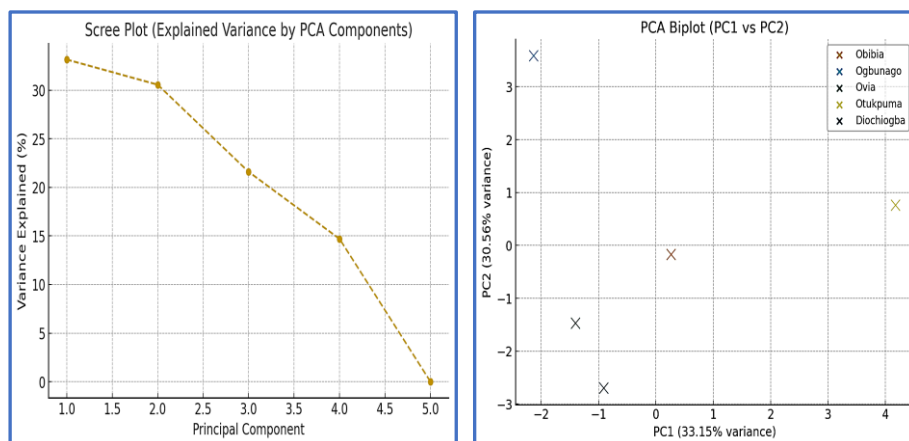


Fig. 4: Principal Component Analysis Plot Showing the Variance Distribution Among the Parameters.

4.3. Water quality index evaluation

The water quality index evaluation in the study was calculated based on the arithmetic weighted analysis of water quality using fifteen (15) geochemical parameters, which has also been used by many authors, and the method has been implemented with 8 – 25 parameters (Shaa-ban et al., 2025; Egun and Oboh, 2022; Egun and Ogiesoba-Eguakun, 2018; and Tyagi et al., 2013), illustrating differences in parameter selection across different studies. (Kumar et al. 2024; Boum-Nkot et al. 2023; Goyal et al. 2023; Das et al. 2022; Makki et al. 2021; Singh et al. 2020). The standard of reference for the parameter in the study was according to the WHO-2022 standards. The diagram (Fig. 5) shows the workflow framework for how the water quality of the study was evaluated diagrammatically.

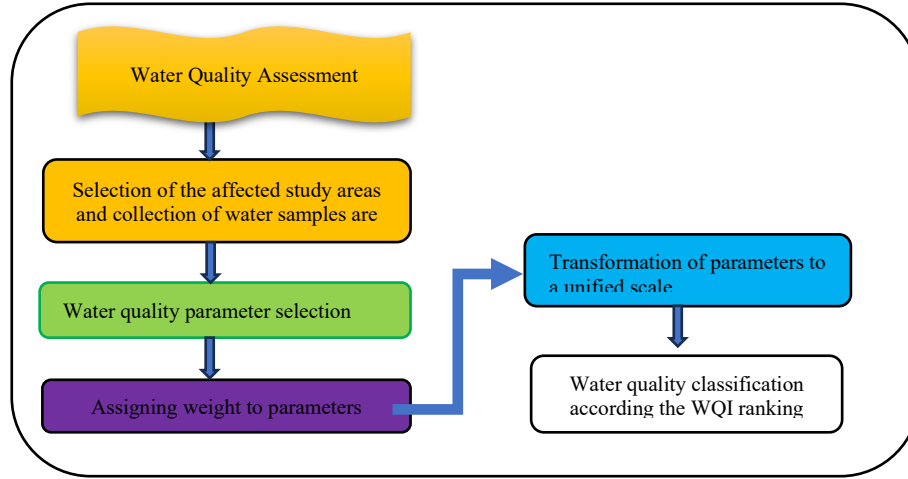


Fig. 5: Flowchart of the Water Quality Assessment of the Study.

The WQI was computed from Eq. 1.

$$WQI = \frac{\sum_{i=1}^n (q_i * W_i)}{\sum_{i=1}^n (W_i)} \quad (1)$$

Where q_i is the quality rating for parameter i and w_i is the unit weight for parameter i ; n represents the number of parameters considered. It is also noted that the unit weight (W_i) is given as

$$W_i = \frac{1}{S_i} \quad (2)$$

Where S_i is the maximum allowable WHO-recommended standard for the i parameter
Also, the quality rating (q_i) is computed as

$$q_i = \frac{C_i}{S_i} * 100 \quad (3)$$

Where C_i is the observed concentration, and S_i is the WHO-recommended standard for the parameter.
The subindex (S_{li}) was also computed using eq 4 as shown below. Where q_i and W_i retain their respective meanings.

$$\text{Subindex } (S_{li}) = q_i * W_i \quad (4)$$

Using the above-listed formula to compute the WQI as presented in Table 2 with respect to the concentration of the physicochemical parameter and WHO-2022 standard of the five water bodies.

The results show that the water in the five rivers is not suitable for human consumption because, when compared to the WQI index rating presented in Table 3, and they are all above 100.

Table 2: The WQI Result of the Five Studied Rivers

Parameters	Otukupuma Rivers	Obibia River	Ogbunago Stream	Ovia Stream	Diochiogba Rivers
EC ($\mu\text{S}/\text{Cm}$)	136.00	120.00	128.00	110.00	115.00
TDS (Mg/L)	32.00	56.00	33.00	32.00	51.00
PH	7.40	7.66	7.52	7.58	4.74
Temperature ($^{\circ}\text{C}$)	30.80	30.00	30.00	30.40	31.00
Turbidity (NTU)	2.40	0.50	0.80	0.30	0.40
Color(pt.co)	3.00	5.00	4.00	2.00	4.00
Hardness (Mg/l)	2100.00	2100.00	1800.00	2000.00	2100.00
BOD(Mg/l)	64.00	70.00	105.00	78.00	100.00
Dissolved oxygen (Mg/l)	16.40	17.20	19.40	15.20	17.80
Nitrate (Mg/l)	0.39	0.23	0.16	0.10	0.11
Chloride (Mg/l)	122.00	138.00	190.00	120.00	135.00
Lead (Mg/l)	0.10	0.14	0.17	0.12	0.10
Iron (Mg/l)	0.50	0.28	0.39	0.20	0.28
Zinc (Mg/l)	0.38	0.40	0.32	0.30	0.42
Arsenic (Mg/l)	0.01	0.02	0.01	0.02	0.01
WQI	542.19	785.55	884.65	687.44	541.78
BIS -WQI	531.68	785.11	868.92	667.16	522.03

Table 3: WQI Rating Table (Brown et al.,1970)

WQI Value	Rating	Grade
0 -25	Excellent	A
26 -50	Good	B
51 -75	Poor	C
76 -100	Very Poor	D
> 100	Unsuitable	E

4.4. BIS-WQI (bureau of Indian standards water quality index)

Similarly, in comparison with the previous WQI approach, the Bureau of Indian Standards Water Quality Index (BIS-WQI) was also used to compare the results and its suitability for human consumption. Using the given parameters, concentration, and Bureau of Indian Standards for water quality, it was observed that there were slight changes in the result of the water quality index as a result of the changes in the Standards used. The results of both analyses, as presented in Table 2, show that the suitability of the rivers for consumption is unsuitable, especially when placed in the classification scale (fig. 5) used by BIS-WQI.

The water WQI was computed using the formula stated below:

$$WQI = \frac{\sum(q_i * W_i)}{\sum W_i} \quad (5)$$

$$q_i = \frac{(V_i - V_{ideal})}{(S_i - V_{ideal})} * 100 \text{ used for pH calculation} \quad (6)$$

$$\text{if } pH \geq 7; q_{pH} = \frac{pH - 7}{8.5 - 7} * 100 \quad (7)$$

$$\text{if } pH \leq 7; q_{pH} = \frac{7 - pH}{7 - 6.5} * 100 \quad (8)$$

Where: V_i - measured value of parameter, S_i - BIS acceptable standard value, V_{ideal} - ideal value; other symbol retain their usual meaning. pH usually has special handling when calculating BIS-WQI; BIS gives an acceptable pH range (6.5 – 8.5) and the ideal value $V_{ideal} = 7.0$, but the other parameter maintains its quality rate (q_i) as

$$q_i = \frac{C_i}{S_i} * 100 \quad (9)$$

The unit weight (W_i) is computed by

$$W_i = \frac{K}{S_i} \quad (10)$$

$$\text{Where } K = \frac{1}{\sum(1/S_i)} \text{ is a proportionality constant} \quad (11)$$

Table 4: BIS-WQI Classification System

< 50	Excellent
50 – 100	Good
100 – 200	Fair
200 – 300	Poor
> 300	Very Poor

The two statistical analyses show similar results in terms of their suitability for water quality. The result of the BIS-WQI shows that the stream/river water body in the study area is not suitable for drinking because its BIS-WQI value was greater than 300 (fig. 5). Both approaches have been successfully used, and there is similarity in the results.

5. Conclusion

The results presented in this study generally highlight the overview of the water quality of five major water bodies in the study area. The physicochemical parameters of the stream/river used in the area were pH, TDS, temperature, turbidity, hardness, conductivity, nitrate, chloride, dissolved oxygen, BOD, color, odor, appearance, arsenic, lead, zinc, and iron. The physicochemical parameters were compared with the WHO standard of the 2022 addendum update. Statistical arithmetic weighted analysis was conducted on the dataset to determine the WQI of the different water bodies; these results were compared with the BIS-WQI for clarity and authenticity. A bar chart was used to determine the distribution of heavy metals, and a correlation plot described the relationship and interaction between each parameter. Then, principal component analysis grouped the dataset into four distinct principal components, which retained their respective original meaning and interpretation. The result of the general WQI of the five rivers/streams shows that all five rivers had a WQI value > 100, which means that the water is not suitable for drinking and other domestic purposes. The result was then compared with the BIS-WQI computational analysis, where BIS-WQI > 300, which means the water was also not suitable for drinking. The results were similar and correlated, and confirmed that the water in the five rivers/streams was not suitable for consumption. Therefore, the government and non-governmental agencies should provide remediation for the citizens within this region, and the citizens using this water should be cautious of water use to prevent future disease outbreaks or cancer.

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